

Nonstop Challenges

In Fault-Tolerant Market

By Omri Serlin

The euphoria of 1980-1983, during which some two dozen start-ups and some well-established companies entered the market for fault-tolerant, on-line transaction processing systems (FT/OLTP), has been replaced recently with the somber realization that selling FT systems is a hard "push" rather than the easy "pull" that entrepreneurs, venture capitalists and some market researchers had anticipated.

With the notable exception of Tandem Computers Inc., Cupertino, Calif., and Stratus Computer Inc., Natick, Mass., no other FT/OLTP suppliers have, as yet, managed the total transition from the development stage to full production.

Just about all new entrants have experienced significant product delays, and many are having a difficult time trying to raise much-needed capital for completing product development and estab-

lishing manufacturing and marketing functions.

Between individual systems, and these are hotly debated among the proponents of the various approaches—"hardware" vs. "software" fault tolerance, tightly coupled vs. loosely coupled architecture, etc.

But in a general sense, just about all FT systems offer reasonably robust fault-detection and recovery mechanisms in two areas: process execution and data storage. The FT requirements of these two aspects are well understood and are relatively easy to achieve.

Unfortunately, processor failures and disk crashes are no longer the key problems preventing nonstop oper-

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penetration by current FT/OLTP suppliers, however, has been minuscule (see chart 2) and is likely to continue to be insignificant in relative terms.

One of the key factors limiting the acceptance of current FT systems is that they generally focus on the easy parts of the problem and tend to ignore the more difficult, and often more important, aspects.

Of course, there are variations be-

tion. There is mounting evidence that "operator errors" and problems with remote communications lines are far more significant factors in downtime. Neither of these problems is effectively addressed in any of today's FT systems.

Furthermore, as the reliability of basic electronic and electro-mechanical components increases, power disruptions are becoming relatively more important in causing failures.

Yet few FT suppliers are providing power-fail protection in their systems. The general attitude seems to be that the provision of reliable power—such as an uninterruptible power system (UPS)—is the responsibility of the customer's facility-management department.

Some very important operational factors are not receiving sufficient attention from FT suppliers. For example, few have solved the problem of how to allow

FT Vendors Facing Difficulty Righting 'Soft' Software Errors

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new versions of the operating system to be installed, without disrupting ongoing operations.

Yet without such a facility, the system can be hardly considered "non-stop." In many cases, the checkpointing (state saving) of the data base, required for protection against disk failures, cannot be accomplished without terminating, or slowing down, the on-line workload.

Some systems do not even provide for on-line repair; such systems cannot be regarded as FT systems at all.

Probably the most difficult issue is that of "soft" software errors, loosely defined as the type of "bugs" in either the system or user code, which do not come to light except under an array of unusual coincidences.

In on-line systems, communications with local and remote terminals often create random sequences that could cause such "bunching" of unusual events.

All these (and other) problems significantly detract from the completeness of the FT story as told by the current practitioners.

Thus, it is not surprising that when the prospective customer is faced with choosing between a "kludge" solution from his conventional supplier and an elegant, but incomplete (and incomplete newcomer without a track record, moreover, often than not the conventional supplier wins.

For example, IBM has been successful in selling its Airline Control Program (ACP) as a high-availability transaction system in the banking and brokerage industries, despite the many severe limitations inherent in the product; NCR Corp., Dayton, Ohio, is now offering its "Cluster" hardware and instant Ready software; and the recent VMS Version 4 from Digital Equipment Corp., Maynard, Mass., implements several high-availability fea-

tures to support its VAXcluster approach.

FT suppliers are responding to the compatibility issue in two ways. Those that hope to place their systems in IBM environments, such as Tandem, are implementing high-level SNA compatibility.

Those aiming at DEC and other non-IBM environments—Auragen Systems Corp., Fort Lee, N.J., Sequoia Systems Inc., Marlboro, Mass., Tolerant Systems Inc., San Jose, Calif.—are providing Unix compatibility. These companies are hoping that the growing acceptance of Unix as a standard will allow them easier entry.

One of the more interesting recent developments is the increasing activity in the area of multi-microprocessor based systems, which offer selected FT features. Arete Systems Corp., Santa Calif., EnMasse Computer Corp., Santa Cruz, Calif., Parallel Computers Inc., Inc., Portland, Ore., and several others are basing their strategies on the idea that most applications currently addressed by "true" FT suppliers could be satisfied with less-than-full fault tolerance, provided the price is right.

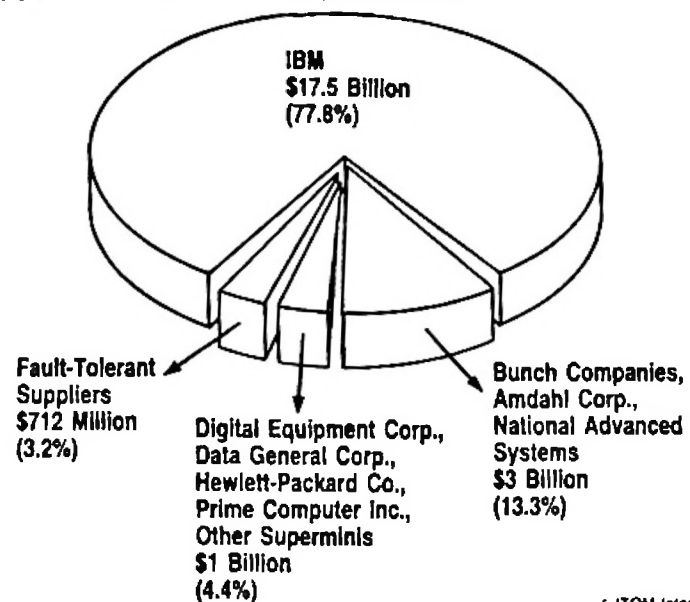
Much attention has been focused on AT&T's recent entry into the fault-tolerant arena with its relatively old 3B20D duplexed system. However, the lack of expandability and of commercial transaction software limits its attraction to defense, telecommunications and other specialized applications.

However, a potentially significant development ensuing from the 3B20 Unix, recently implemented by AT&T Bell Laboratories, that allow it to run on the dual-processor 3B20A model. These modifications, which are expected to eventually show up in Unix System V, are general enough to allow Unix to be used on other multiple processor configurations.

1984 Transaction Processing Market (Projections)

Chart 1

TOTAL: \$22.5 Billion

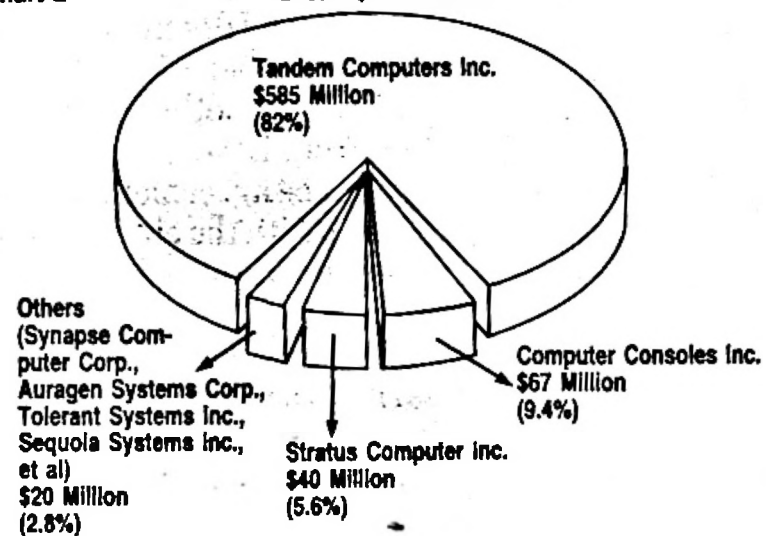


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1984 Penetration Of On-Line Transaction Processing Market By Fault-Tolerant Suppliers (Projections)

Chart 2

TOTAL: \$712 Million



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Fault-Tolerant Solutions Vary With Vendor

By Paul E. Schindler Jr.

Every commercial fault-tolerant computer system uses some combination of both hardware and software to ensure that the systems will crash less often

than conventional computer systems when component failures occur.

Still, the various fault-tolerant systems suppliers fall at various places along a spectrum

that runs between hardware solutions and software solutions, vendors and industry analysts said. Most vendors combine the two approaches to some degree. At the hardware end of the

spectrum are those who believe that fault tolerance should rely on CPU redundancy. On the other end are those who said it should rely on software to isolate component or software failures.

Senior analyst Joan de Regt of International Resource Development Inc., a Norwalk, Conn., marketing research firm, said vendors depending primarily on hardware for fault tolerance include August Systems Inc., Tigard, Ore., which uses a triple-CPU checking system; Stratus Computer Inc., Natick, Mass., which has four CPUs checking each transaction; and AT&T, which relies on self-checking VLSI chips.

"Hardware-based systems are more fault tolerant than the software-based systems because you can immediately redirect traffic to the working components," said de Regt.

Hardware solutions, said Shirley Henry, director of marketing at Tolerant Systems Inc., San Jose, Calif., are best when absolute system integrity at the instruction level is needed. "If a human life is at stake, use redundant hardware," she said. However, Henry said it was "fair to say we are primarily using software" to achieve fault tolerance.

She, and others who build software-based systems, believe that such systems are more easily expandable than hardware-based systems, since they rely on reconfiguration rather than duplication of components.

Software solutions may still be virtually instantaneous in recovery time, but usually several seconds are needed to reconfigure the system, industry analysts said.

Among those offering software solutions are Tandem Computers Inc., Cupertino, Calif., the founder of the fault-tolerant marketplace; Auragen Systems Corp., Englewood Cliffs, N.J.; Computer Consoles Inc., Rochester, N.Y.; Parallel Computers, Santa Cruz, Calif.; Sequoia Systems Inc., Marlboro, Mass.; Synapse Computer Corp., Milpitas, Calif.; and Tolerant Systems.

Many of these software-dependent, fault-tolerant systems are based on Motorola's MC 68000 32-bit chip, and have Unix-compatible or enhanced Unix operating systems.

Tandem, however, combines software fault tolerance with a duplicate CPU for backup operation.

Tolerant, like fault-tolerant vendor Synapse, has broken its CPU into segments of logic and memory that can distribute a computation around a failure.

Tandem, the market leader with more than \$500 million in sales this year, relies on both software fault tolerance, which is customized for each user, and on hardware redundancy provided by a backup CPU.

De Regt said it was her opinion that Tandem's philosophy of

software-dependent fault tolerance made sense when the firm was founded in the mid-70s, "at a time when hardware costs were high and software was less expensive."

Ever since, however, hardware costs have dropped and software costs have increased, she said, "so the balance has shifted," making hardware solutions to fault tolerance more effective.

Charles Lecht, president of Lecht Sciences Inc., a New York software house, had a less even-handed approach to the question than de Regt. "Redundant processors are the only way to go," he said. "All of them running the same program, all of them able to take over each other's workload."

"Of course, software will always be a factor," Lecht said. "But if you think of software as the driver and hardware as the car, all the drivers in the world will not help you if there is no car for them to drive. You cannot do it all in software."

The philosophical leader of the hardware fault-tolerant camp, analysts agree, is Stratus. Peter Kastner, manager of Stratus' corporate business development, said his 4-year-old firm believes that hardware fault tolerance has "definite advantages" over software-based systems.

Competitors argue that redundant hardware systems such as Stratus—which has four sets of logic executing the same instructions simultaneously—has a cost disadvantage. But Kastner said chips amount to only 2 percent of the selling price of a typical system. The clear advantage is that any malfunctioning chip—the one that produces a different answer than the others—is immediately isolated from the system for repairs. Moreover, he said, hardware faults are far more infrequent than software faults.

Kastner summed up the Stratus philosophy as "hardware redundancy which is controlled by hardware, rather than by software," and argued that his firm's software techniques, in-

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cluding automatic rebooting in case of system failure, use fewer CPU cycles than similar techniques used by software-oriented competitors.

Toward the hardware end of

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the spectrum, but less militant than Stratus, is AT&T Technologies, formerly the Western Electric Co. subsidiary of AT&T. Paul Teetor, an engineer with AT&T Bell Laboratories in Naperville, Ill., said that while AT&T's 3B20D processor does offer hardware redundancy, "our strength is the fusion of hardware and software."

Teetor admitted that "hardware redundancy is the backbone of our system," but said it would not offer the benefits it does without use of a special version of the Unix operating system known as Unix Real Time Reliable or Unix RTR.

When AT&T announced commercial availability of the 3B20D last March, Teetor said, it said a typical system operating 24 hours a day, 365 days a year would only be down eight minutes a year. When the 3B20D was being designed, tested and used within AT&T's phone system, it was responsible for keeping the phone system's 800 numbers and AT&T credit-card verification services up and running.

Previously written Unix programs will run under Unix RTR on the 3B20D without modification, Teetor said, and they will be more reliable because they are running on a more reliable operating system or can be modified to take advantage of the system's special fault-tolerant features.

"Our whole goal is to lose the minimum amount of information during a failure," he said. As a result, the AT&T system, unlike Stratus, does not immediately have an automatic reboot feature. Instead, it first tries to restart the process, using a combination of hardware and software to prevent entry into infinite loops.

Hardware systems are criticized for the difficulty in expanding them, due to the need for hardware overhead to provide redundancy. By contrast, software systems are more readily expandable, since the amount of software needed does not grow as quickly as does system size.

Tandem, which relies on both software fault tolerance and CPU duplication, prefers to think of fault tolerance as "a feature, not a market," according to Jerry Held, Tandem's director of strategic planning.

Held also said that Tandem's self-view is not that of a fault-tolerant system provider, but rather a provider of systems for "on-line transaction processing," for which fault tolerance is but one requirement.

His company, founded in 1970, is the market leader, with more than 80 percent of the market.

Held said he prefers to look at a different spectrum, "not from hardware to software, but from availability to expandability." Most systems fall at one end or the other of this scale, he said, while Tandem is firmly placed in the middle, "offering the best combination of both."

Single-processor redundant systems such as Stratus and AT&T would be at the 100 percent-availability end of the scale, he suggested, while Synapse has stressed expandability over 100 percent uptime.

But, while saying that Tandem systems offer "a tremendous amount of hardware fault tolerance," Held was willing to say that "a number of our features are based in software." He said Tandem has examined the concept of redundant hardware, but concluded that "it is not optimum."

Hardware solutions, Held said, "only solve half the problem. We believe multiprocessor systems are the key to most customer problems." Such systems are easier to put together with software than with hardware, he said.

While the cost of hardware components is dropping, Held said, the number of components in a system is rising. "Hardware cost is not going to zero," he said. "Redundant systems tend to double hardware costs. In two systems, with equally mature technologies, the more cost-effective design will be software-oriented."

Users Find 'Fault Tolerance'

A Relative Term

By Jean S. Bozman

Fault tolerance is a relative term end users told *Information Systems News*. For some, a short time period for system recovery is acceptable, while for others nothing less than non-stop processing will do.

Fault tolerance is a phrase that is thrown around, said one Dallas user, "but nobody knows exactly what it means. A single hardware failure may not crash a system, but vital data may be lost during an incomplete transaction. And, in any case, software failures are more often the causes of user downtime than hardware failures."

Because fault-tolerant systems vary so greatly in real-time transaction processing, some users have gone to great lengths to test the limits of systems they plan to purchase. One company, a New York investment firm, tested a Stratus Computer Inc. system by seeing how many printed boards could be removed without a system failure.

"We found that the Stratus really does tolerate a lot of stress," said an information systems manager at the New York firm. "We started pulling printed-circuit boards at random, and the only way we got it to stop was to pull out enough CPU boards to make it non-functional. However, the moment we replaced the CPU boards, it was off and running again."

That kind of real-time transaction processing—the kind used in process control, banking and financial applications—puts at a premium the 100-percent availability of the computer.

For many applications, sub-second or several-second recovery time is sufficient, users and vendors agreed.

"In an office situation, where you have a lot of people sitting at terminals, users will tolerate a several-second delay for recovery far better than a bank that is doing hundreds of financial transactions each minute," said one engineer at AT&T who did not wish to be named. "Engi-

neering environments demand immediate response, as do telecommunications applications."

Tolerable waiting times vary widely, vendors said. "The amount of acceptable recovery time is application-dependent," said Shirley Henry, director of marketing at Tolerant Systems Inc., San Jose, Calif. "The level of protection is selectable by programming."

To achieve this, Tolerant Systems relies on a software solution, allowing an error-detector buried in the "kernel" of its enhanced Unix operating system to isolate hardware or software failures that result in an incomplete transaction, bypassing them until repairs can be made. Then, the system reconfigures around the problem, with recovery time in a second or two.

"Our system, which is made up of system building blocks based on the National Semiconductor [Corp.] 16032 and 32032 chips, will not allow a partial transaction to take place," Henry said. "If a stoppage occurs, the computer will return to a previous copy of the data base and reconstruct the transaction from the beginning." The integrity of the data base in a system

such as Tolerant's is preserved. However, a certain amount of time is needed before the user can then access the data base.

Hardware-based, fault-tolerant systems offer an additional advantage, analysts point out—

the data base was never disturbed by the failure of one of the CPUs or its components.

Information systems managers should be analyzing what they really want in a fault-tolerant system, said the AT&T engineer. "When

such a decision-maker buys fault tolerance, he's really buying faith," the engineer said. "He knows there will inevitably be hardware failures. But the real question is will the machine keep running even when those failures occur?"